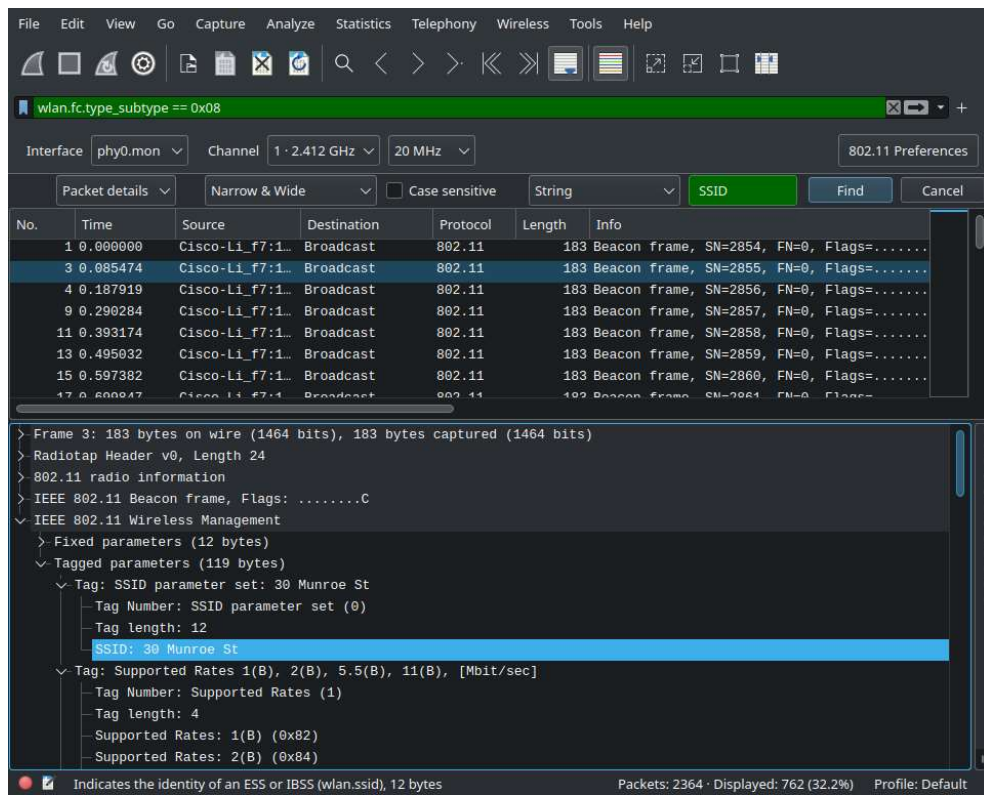
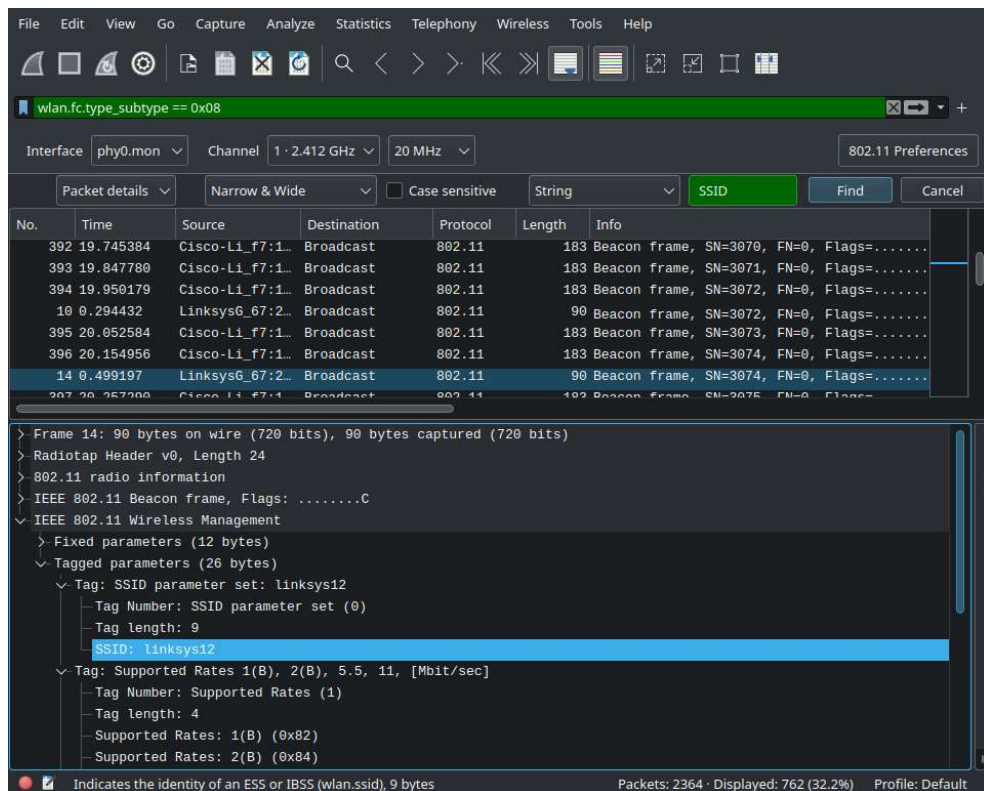


WIRELESS

1. Koji su SSID oznake dvije pristupne točke koje su poslale većinu beacon okvira?

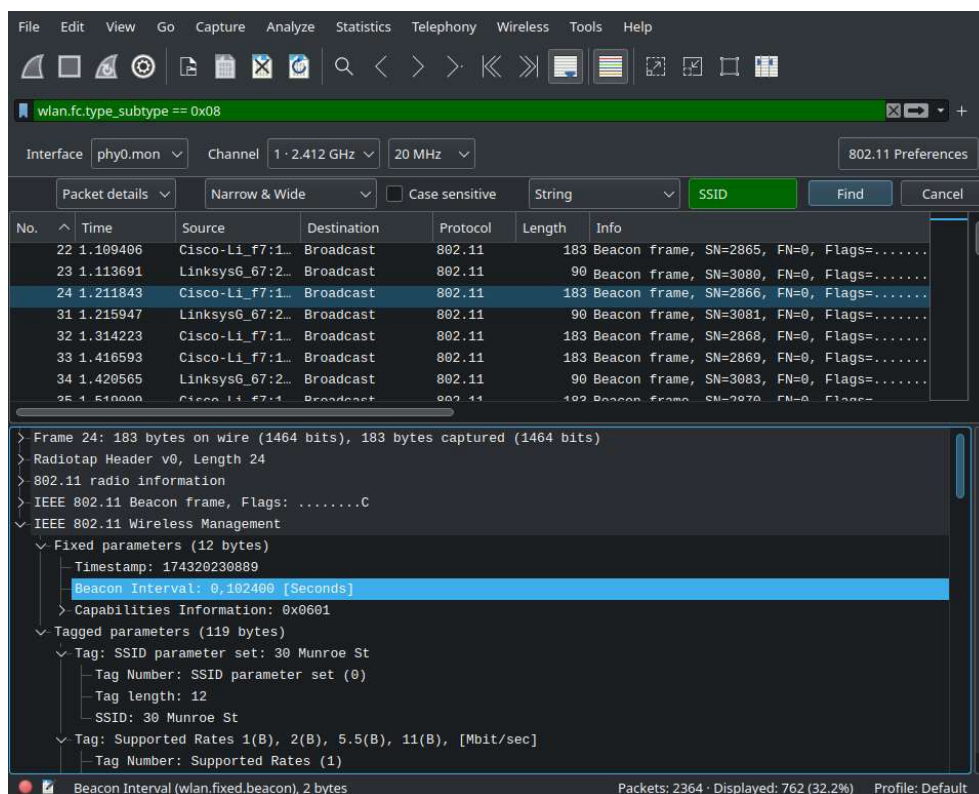


Slika 1 – SSID oznake 1

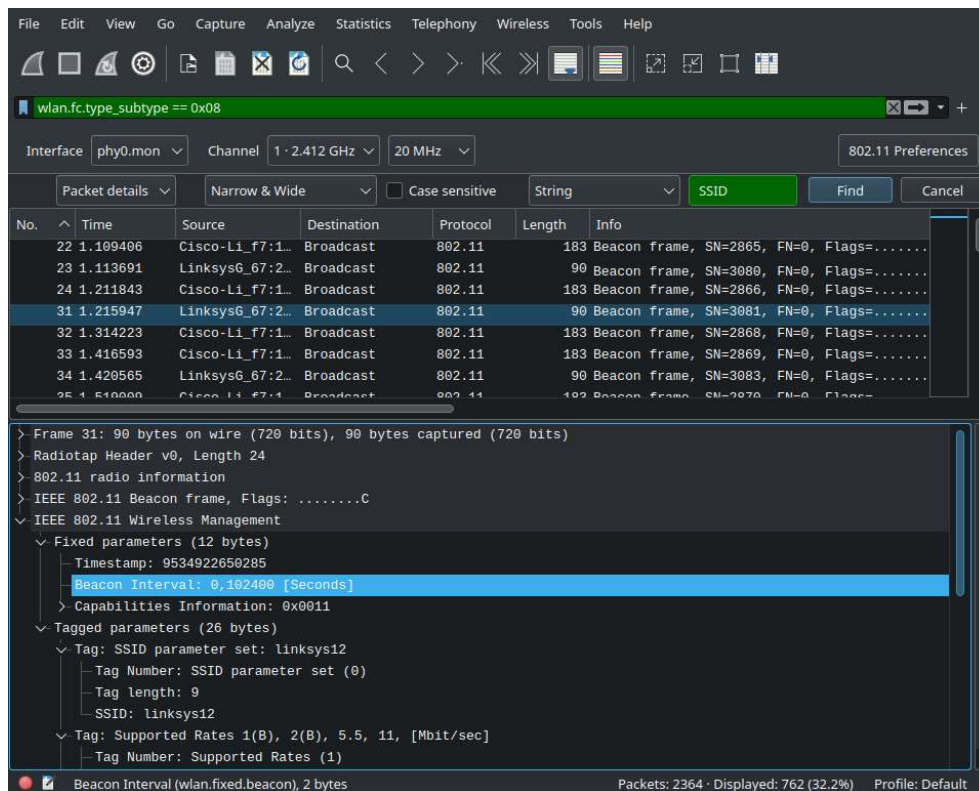


Slika 2 – SSID oznake 2

2. Koliki je vremenski interval između dva slanja beacon okvira za linksys12 pristupnu točku, a koliki za „30 Munroe St“ pristupnu točku?

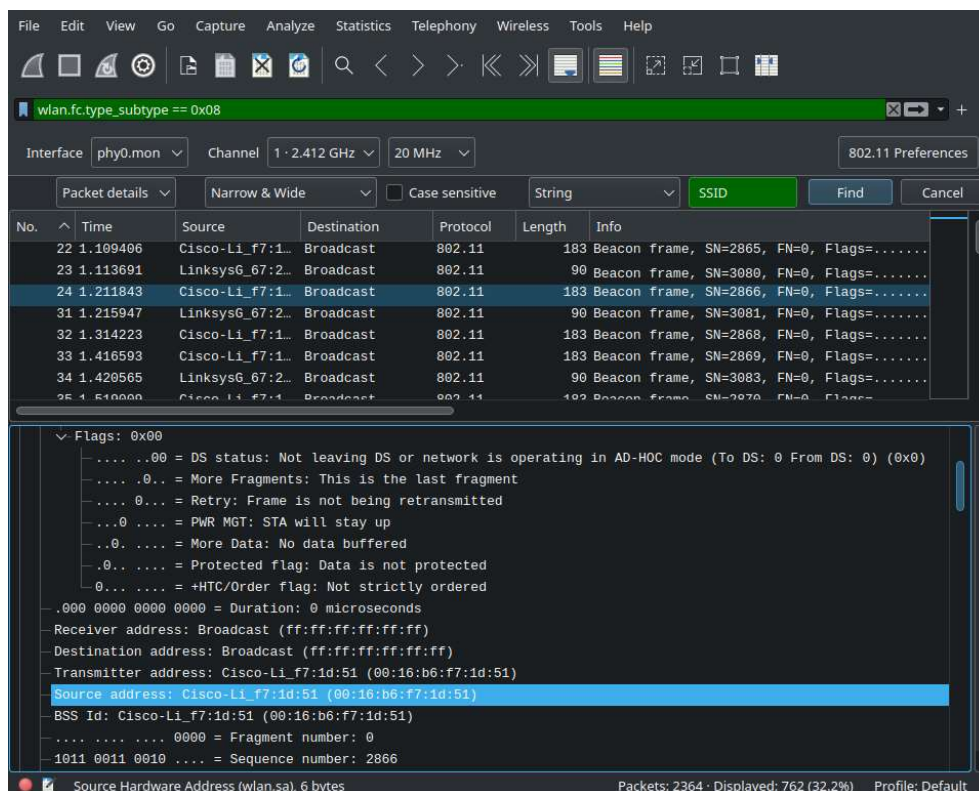


Slika 3 – vremenski interval 1



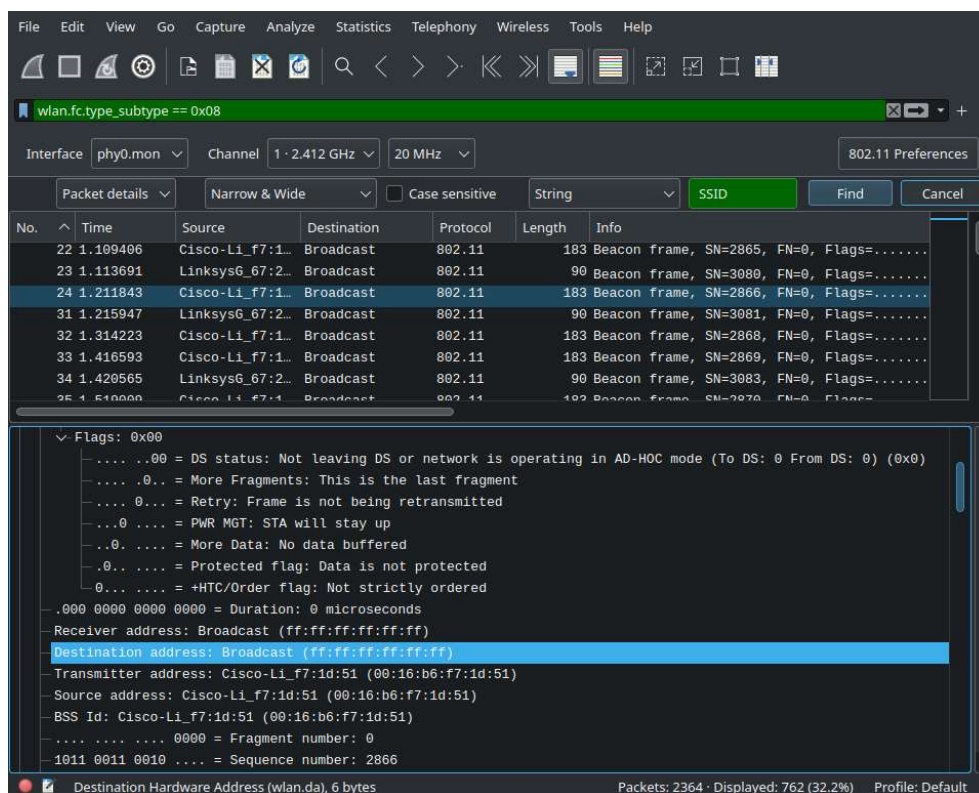
Slika 4 – vremenski interval 2

3. Koja je izvorišna MAC adresa beacon okvira sa 30 Munroe St?



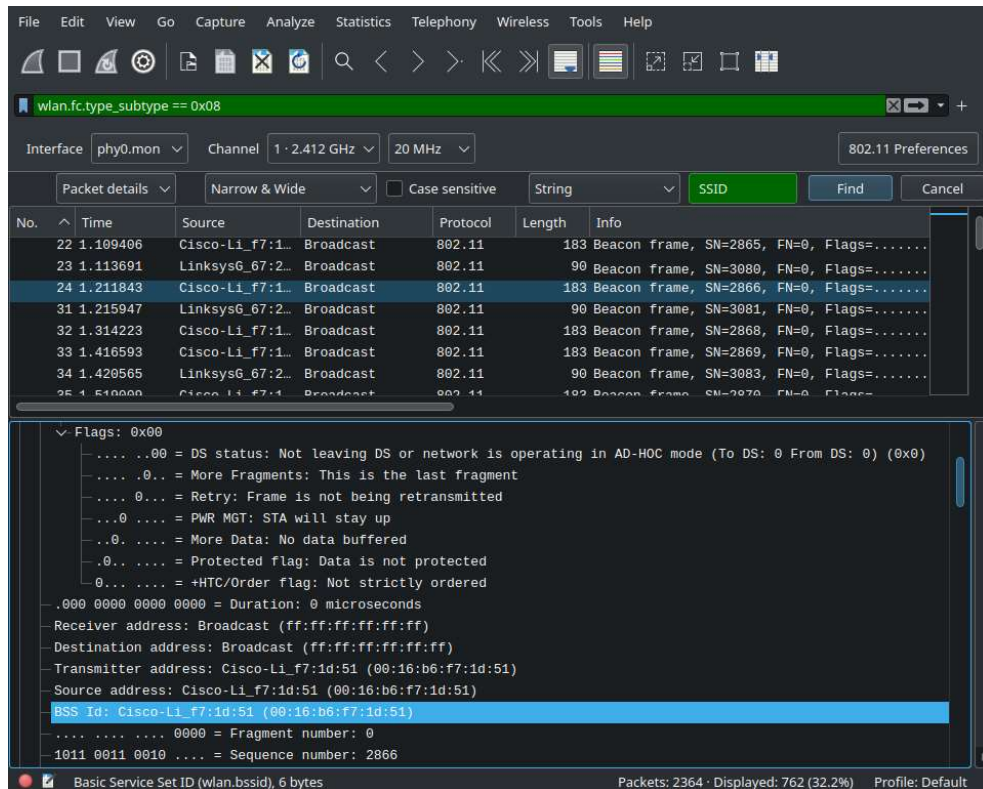
Slika 5 – izvorišna MAC adresa

4. Koja je odredišna MAC adresa beacon okvira sa 30 Munroe St?



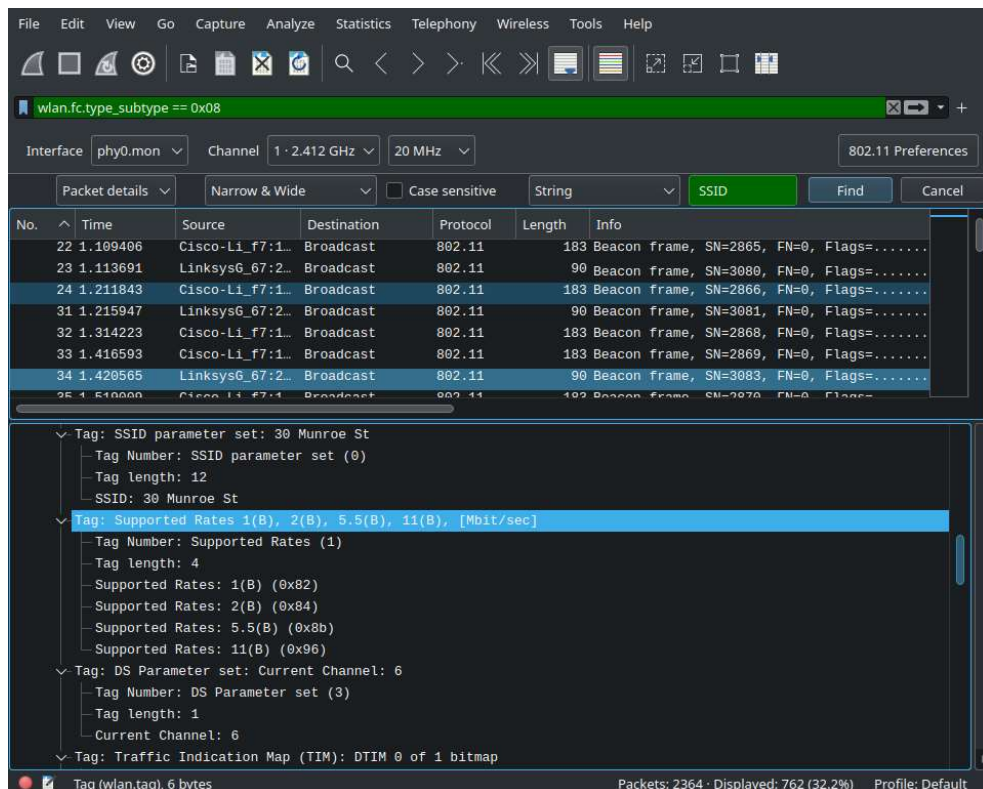
Slika 6 – odredišna MAC adresa

5. Koja je BSS Id MAC adresa beacon okvira sa 30 Munroe St?

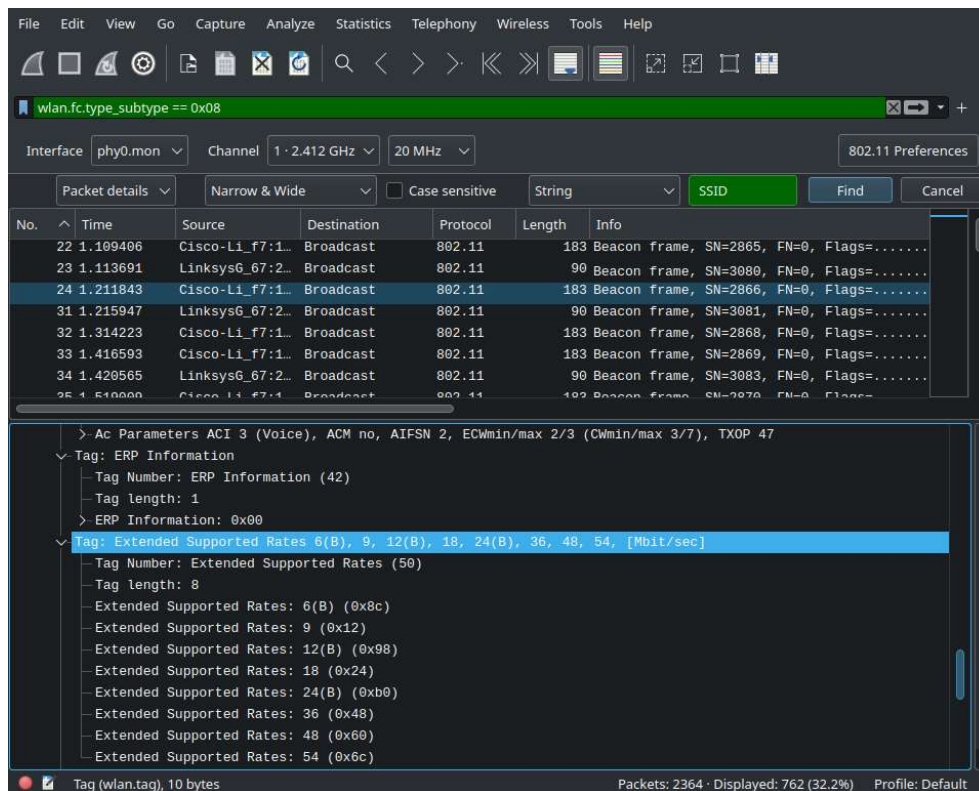


Slika 7 – BSS Id MAC adresa

6. Beacon okvir sa 30 Munroe St pristupne točke oglašava da podržava četiri brzine prijenosa i osam dodatnih “extended supported rates.” O kojim se brzinama radi?



Slika 8 – brzine prijenosa koje se podupiru



Slika 9 - extended supported brzine prijensa

7. Pronađite 802.11 okvir koji sadrži SYN TCP segment za prvu TCP vezu (kojom se snima alicex.txt). Koje su tri MAC adrese u poljima zaglavlja 802.11 okvira? Koja MAC adresa pripada računalu s bežičnom vezom, koja pristupnoj točki, a koja usmjerniku „prvog skoka“ (koji povezuje kućnu mrežu sa Internetom)? Koja je IP adresa računala s bežičnom mrežom koje šalje prvi TCP segment? Koja je odredišna IP adresa? Pripada li odredišna IP adresa računalu s bežičnom vezom, pristupnoj točki ili usmjerniku „prvog skoka“? Obrazložite!

Podaci o MAC adresama se mogu očitati u IEEE 802.11 Qos Data

IntelCor_d1:b6:4f (00:13:02:d1:b6:4f) – Računalo za bežičnom vezom - **Source** address

Cisco-Li_f7:1d:51 (00:16:b6:f7:1d:51) – pristupna točka – **Receiver** address

Cisco-Li_f4:eb:a8 (00:16:b6:f4:eb:a8) – usmjernik prvog skoka – **Destination** address

IP adresa računala s bežičnom mrežom: 192.168.1.109

Odredišna IP adresa: 128.119.245.12

Odredišna IP adresa `128.119.245.12` pripada vanjskom serveru (`gaia.cs.umass.edu`), a ne računalu s bežičnom vezom, pristupnoj točki ili usmjerniku "prvog skoka".

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn == 1

Interface phy0.mon Channel 1 · 2.412 GHz 20 MHz 802.11 Preferences

No.	Time	Source	Destination	Protocol	Length	Info
474	24.811093	192.168.1.109	128.119.245.12	TCP	110	2538 → 80 [SYN] Seq=0 Win=16384 Len=0 MSS=
476	24.827751	128.119.245.12	192.168.1.109	TCP	110	80 → 2538 [SYN, ACK] Seq=0 Ack=1 Win=5840
1011	32.808574	192.168.1.109	128.119.240.19	TCP	110	2541 → 80 [SYN] Seq=0 Win=16384 Len=0 MSS=
1013	32.825631	128.119.240.19	192.168.1.109	TCP	110	80 → 2541 [SYN, ACK] Seq=0 Ack=1 Win=5840
1034	32.869262	192.168.1.109	128.119.240.19	TCP	110	2542 → 80 [SYN] Seq=0 Win=16384 Len=0 MSS=
1047	32.890900	128.119.240.19	192.168.1.109	TCP	110	80 → 2542 [SYN, ACK] Seq=0 Ack=1 Win=5840
1048	32.890998	128.119.240.19	192.168.1.109	TCP	110	[TCP Out-Of-Order] 80 → 2542 [SYN, ACK] Se
1051	32.891536	128.119.240.19	192.168.1.109	TCP	110	[TCP Out-Of-Order] 80 → 2542 [SYN, ACK] Se
1058	32.903185	128.119.101.5	192.168.1.109	TCP	110	80 → 2543 [SYN, ACK] Seq=0 Ack=1 Win=5840
1119	32.957207	192.168.1.109	128.119.240.19	TCP	110	2544 → 80 [SYN] Seq=0 Win=16384 Len=0 MSS=

> Frame 474: 110 bytes on wire (880 bits), 110 bytes captured (880 bits)

> Radiotap Header v0, Length 24

> 802.11 radio information

IEEE 802.11 QoS Data, Flags:TC

- Type/Subtype: QoS Data (0x0028)
- Frame Control Field: 0x8801
- Duration: 44 microseconds
- Receiver address: Cisco-Li_f7:1d:51 (00:16:b6:f7:1d:51)
- Transmitter address: IntelCor_d1:b6:4f (00:13:02:d1:b6:4f)
- Destination address: Cisco-Li_f4:eb:a8 (00:16:b6:f4:eb:a8)
- Source address: IntelCor_d1:b6:4f (00:13:02:d1:b6:4f)
- BSS Id: Cisco-Li_f7:1d:51 (00:16:b6:f7:1d:51)
- STA address: IntelCor_d1:b6:4f (00:13:02:d1:b6:4f)
- Fragment number: 0
- Sequence number: 49

wireless1.pcap Packets: 2364 · Displayed: 24 (1.0%) Profile: Default

Slika 10 – MAC adrese

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn == 1 and tcp.flags.ack == 1

Interface phy0.mon Channel 1 · 2.412 GHz 20 MHz 802.11 Preferences

No.	Time	Source	Destination	Protocol	Length	Info
476	24.827751	128.119.245.12	192.168.1.109	TCP	110	80 → 2538 [SYN, ACK] Seq=0 Ack=1 Win=5840
1013	32.825631	128.119.240.19	192.168.1.109	TCP	110	80 → 2541 [SYN, ACK] Seq=0 Ack=1 Win=5840
1047	32.890900	128.119.240.19	192.168.1.109	TCP	110	80 → 2542 [SYN, ACK] Seq=0 Ack=1 Win=5840
1048	32.890998	128.119.240.19	192.168.1.109	TCP	110	[TCP Out-Of-Order] 80 → 2542 [SYN, ACK] Se
1051	32.891536	128.119.240.19	192.168.1.109	TCP	110	[TCP Out-Of-Order] 80 → 2542 [SYN, ACK] Se
1058	32.903185	128.119.101.5	192.168.1.109	TCP	110	80 → 2543 [SYN, ACK] Seq=0 Ack=1 Win=5840
1130	32.978707	128.119.240.19	192.168.1.109	TCP	110	80 → 2544 [SYN, ACK] Seq=0 Ack=1 Win=5840
1134	32.979160	128.119.240.19	192.168.1.109	TCP	110	80 → 2545 [SYN, ACK] Seq=0 Ack=1 Win=5840
1165	33.021747	128.119.240.19	192.168.1.109	TCP	110	80 → 2547 [SYN, ACK] Seq=0 Ack=1 Win=5840
1177	33.030503	64.233.187.104	192.168.1.109	TCP	108	80 → 2546 [SYN, ACK] Seq=0 Ack=1 Win=8190

> Frame 476: 110 bytes on wire (880 bits), 110 bytes captured (880 bits)

> Radiotap Header v0, Length 24

> 802.11 radio information

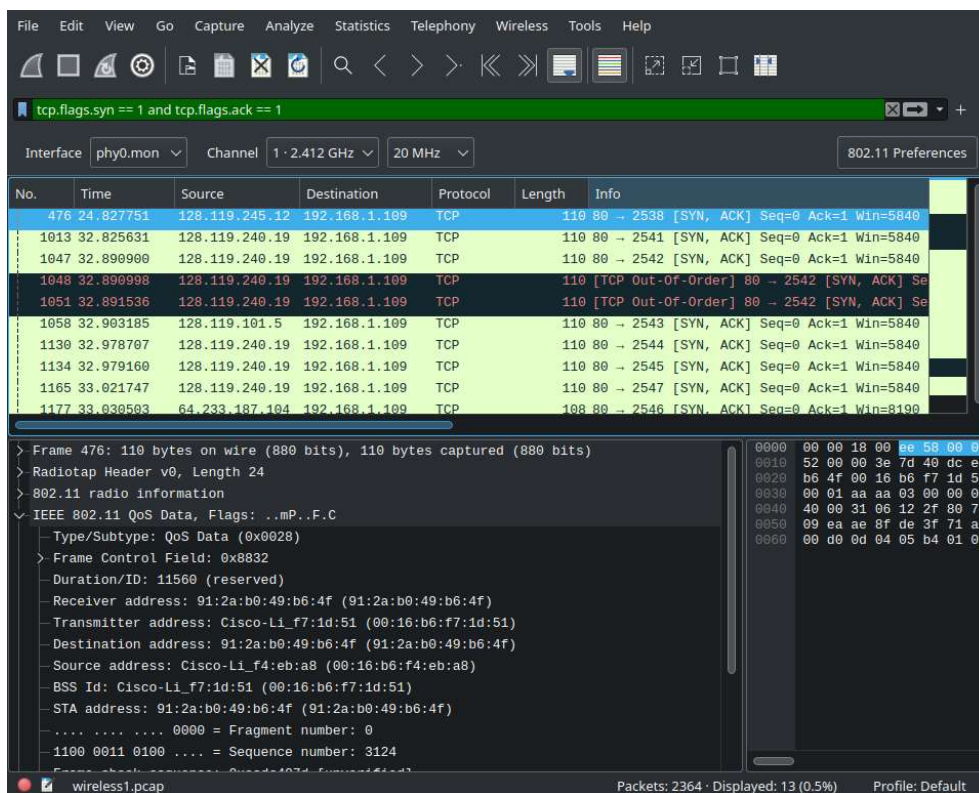
IEEE 802.11 QoS Data, Flags: ..mP..F.C

- Type/Subtype: QoS Data (0x0028)
- Frame Control Field: 0x8832
- Duration/ID: 11560 (reserved)
- Receiver address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
- Transmitter address: Cisco-Li_f7:1d:51 (00:16:b6:f7:1d:51)
- Destination address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
- Source address: Cisco-Li_f4:eb:a8 (00:16:b6:f4:eb:a8)
- BSS Id: Cisco-Li_f7:1d:51 (00:16:b6:f7:1d:51)
- STA address: 91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f)
- Fragment number: 0
- Sequence number: 3124

wireless1.pcap Packets: 2364 · Displayed: 13 (0.5%) Profile: Default

Slika 11 – MAC adrese

8. Pronađite 802.11 okvir koji sadrži SYNACK segment za istu TCP vezu. Koje su tri MAC adrese u tom 802.11 okviru? Koja MAC adresa pripada računalu s bežičnom vezom, koja pristupnoj točki, a koja usmjerniku „prvog skoka“ (koji povezuje kućnu mrežu sa Internetom)? Jesu li MAC adrese pošiljatelja okvira i IP adresa pošiljatelja datagrama u kojem se nalazi TCP segment adrese istog uređaja? Objasni!



Slika 12 – SYN ACK

Ista TCP veza znači sada put slanja se mijenja Destination u Source

91:2a:b0:49:b6:4f (91:2a:b0:49:b6:4f) - Računalo za bežičnom vezom – Destination address

Cisco-Li_f7:1d:51 (00:16:b6:f7:1d:51) - pristupna točka – **Transmitter** address

Cisco-Li_f4:eb:a8 (00:16:b6:f4:eb:a8) – usmjernik prvog skoka - Source address

MAC adrese pošiljatelja okvira i IP adresa pošiljatelja datagrama u kojem se nalazi TCP segment nisu adrese istog uređaja. MAC adresa 00:16:b6:f4:eb:a8 pripada **usmjerniku prvog skoka** IP adresa 128.119.245.12 pripada računalu **gaia.cs.umass.edu**.

9. Koje akcije poduzima (koje okvire šalje) računalo da bi prekinulo vezu sa „30 Munroe St“ pristupnom točkom (vrijeme t=49)? (Pomoć: prvo je akcija IP-sloja, a nakon toga ide akcija 802.11-sloja)

Paket 1733: bežični host šalje zahtjev za oslobodjenje dodijeljenih IP adresa - **DHCP Release**

No.	Time	Source	Destination	Protocol	Length	Info
1727	49.429849	Intel_d1:b6:4f	CiscoLinksys_f7:1d:...	802.11	54	QoS Null function (No data), SN=...
1728	49.430007	Intel_d1:b6:4f	Intel_d1:b6:4f (00:...	802.11	38	Acknowledgement, Flags=.....C
1729	49.440041	CiscoLinksys_f7:1d:...	Broadcast	802.11	183	Beacon frame, SN=3587, FN=0, Fla...
1730	49.440146	Intel_d1:b6:4f	CiscoLinksys_f7:1d:...	802.11	54	QoS Null function (No data), SN=...
1731	49.440243	Intel_d1:b6:4f	Intel_d1:b6:4f (00:...	802.11	38	Acknowledgement, Flags=.....C
1732	49.542481	CiscoLinksys_f7:1d:...	Broadcast	802.11	183	Beacon frame, SN=3588, FN=0, Fla...
1733	49.583615	192.168.1.109	192.168.1.1	DHCP	390	DHCP Release - Transaction ID 0...
1734	49.583771	Intel_d1:b6:4f	Intel_d1:b6:4f (00:...	802.11	38	Acknowledgement, Flags=.....C
1735	49.609617	Intel_d1:b6:4f	CiscoLinksys_f7:1d:...	802.11	54	Deauthentication, SN=1605, FN=0, F...
1736	49.609770	Intel_d1:b6:4f	Intel_d1:b6:4f (00:...	802.11	38	Acknowledgement, Flags=.....C
1737	49.614478	Intel_d1:b6:4f	Broadcast	802.11	99	Probe Request, SN=1606, FN=0, Fl...
1738	49.615869	CiscoLinksys_f5:ba:...	CiscoLinksys_f5:ba:...	802.11	38	Acknowledgement, Flags=.....C
1739	49.617713	CiscoLinksys_f5:ba:...	CiscoLinksys_f5:ba:...	802.11	38	Acknowledgement, Flags=.....C
1740	49.638857	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	58	Authentication, SN=1606, FN=0, F...
1741	49.639700	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	58	Authentication, SN=1606, FN=0, F...

<

> Frame 1733: 390 bytes on wire (3120 bits), 390 bytes

> Radiotap Header v0, Length 24

> 802.11 radio information

> IEEE 802.11 QoS Data, Flags:TC

> Logical-Link Control

> Internet Protocol Version 4, Src: 192.168.1.109, Dst: 192.168.1.1

> User Datagram Protocol, Src Port: 68, Dst Port: 67

> Dynamic Host Configuration Protocol (Release)

0000

00 00 18 00 ee 58 00 00 10 6c 85 09 c0 00 e5 9c

0010

60 00 00 49 91 17 38 90 88 01 2c 00 00 16 b6 f7

0020

1d 51 00 13 02 d1 b6 4f 00 16 b6 f4 eb a8 80 0b

0030

00 00 aa aa 03 00 00 00 08 00 45 00 01 48 14 1c

0040

00 00 80 11 a1 ca c0 a8 01 6d c0 a8 01 01 00 44

0050

00 43 01 34 5e d8 01 01 06 00 0e a5 a5 26 04 00

0060

80 00 c0 a8 01 6d 00 00 00 00 00 00 00 00 00 00

0070

00 00 00 13 02 d1 b6 4f 00 00 00 00 00 00 00 00

0080

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0090

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Slika 13 - DHCP Release

Paket 1735: oslobađa se IP adresa koju je DHCP server dodijelio hostu – **Deauthentication**

wireless1.pcap

File

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Wireless

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+

No.	Time	Source	Destination	Protocol	Length	Info
1727	49.429849	Intel_d1:b6:4f	CiscoLinksys_f7:1d:...	802.11	54	QoS Null function (No data), SN=...
1728	49.430007	Intel_d1:b6:4f	Intel_d1:b6:4f (00:...	802.11	38	Acknowledgement, Flags=.....C
1729	49.440041	CiscoLinksys_f7:1d:...	Broadcast	802.11	183	Beacon frame, SN=3587, FN=0, Fla
1730	49.440146	Intel_d1:b6:4f	CiscoLinksys_f7:1d:...	802.11	54	QoS Null function (No data), SN=...
1731	49.440243	Intel_d1:b6:4f	Intel_d1:b6:4f (00:...	802.11	38	Acknowledgement, Flags=.....C
1732	49.542481	CiscoLinksys_f7:1d:...	Broadcast	802.11	183	Beacon frame, SN=3588, FN=0, Fla
1733	49.583615	192.168.1.109	192.168.1.1	DHCP	390	DHCP Release - Transaction ID 0
1734	49.583771	Intel_d1:b6:4f	Intel_d1:b6:4f (00:...	802.11	38	Acknowledgement, Flags=.....C
1735	49.609617	Intel_d1:b6:4f	CiscoLinksys_f7:1d:...	802.11	54	Deauthentication, SN=1605, FN=0,
1736	49.609770	Intel_d1:b6:4f	Intel_d1:b6:4f (00:...	802.11	38	Acknowledgement, Flags=.....C
1737	49.614478	Intel_d1:b6:4f	Broadcast	802.11	99	Probe Request, SN=1606, FN=0, Fl
1738	49.615869	CiscoLinksys_f5:ba:...	CiscoLinksys_f5:ba:...	802.11	38	Acknowledgement, Flags=.....C
1739	49.617713	CiscoLinksys_f5:ba:...	CiscoLinksys_f5:ba:...	802.11	38	Acknowledgement, Flags=.....C
1740	49.638857	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	58	Authentication, SN=1606, FN=0, F
1741	49.639700	Intel_d1:b6:4f	CiscoLinksys_f5:ba:...	802.11	58	Authentication, SN=1606, FN=0, F

<

>

> Frame 1735: 54 bytes on wire (432 bits), 54 bytes cap

> Radiotap Header v0, Length 24

> 802.11 radio information

> IEEE 802.11 Deauthentication, Flags:C

Type/Subtype: Deauthentication (0x000c)

Frame Control Field: 0xc000

.000 0000 0010 1100 = Duration: 44 microseconds

Receiver address: CiscoLinksys_f7:1d:51 (00:16:b6:

Destination address: CiscoLinksys_f7:1d:51 (00:16:

Transmitter address: Intel_d1:b6:4f (00:13:02:d1:b

Source address: Intel_d1:b6:4f (00:13:02:d1:b6:4f)

BSS Id: CiscoLinksys_f7:1d:51 (00:16:b6:f7:1d:51)

.... .. 0000 = Fragment number: 0

0110 0100 0101 = Sequence number: 1605

Frame check sequence: 0x3b4a8b9c [unverified]

[FCS Status: Unverified]

[WLAN Flags:C]

> IEEE 802.11 Wireless Management

0000 00 00 18 00 ee 58 00 00 10 6c 85 09 c0 00 e5 9c

0010 55 00 00 49 9c 8b 4a 3b c0 00 2c 00 00 16 b6 f7

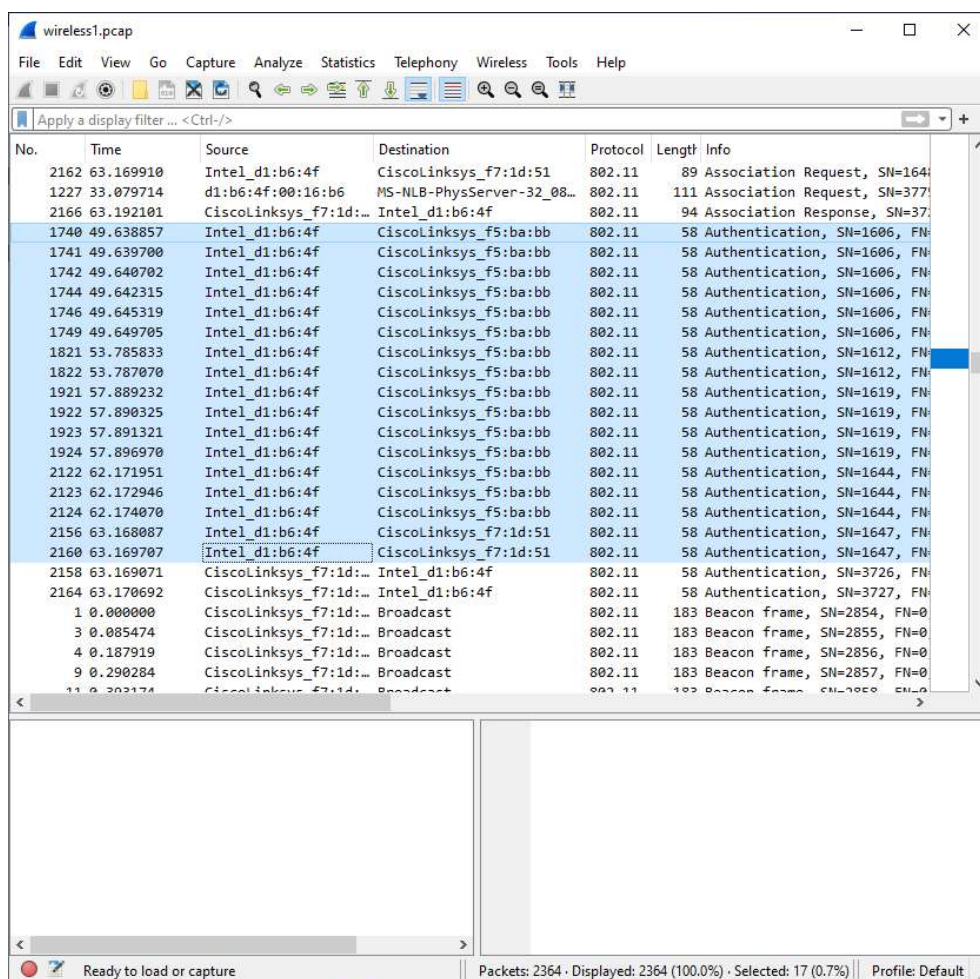
0020 1d 51 00 13 02 d1 b6 4f 00 16 b6 f7 1d 51 50 64

0030 01 00 9c 8b 4a 3b

Slika 14 - Deauthentication

10. Pregledajte AUTHENTICATION okvire koje je računalo poslalo pristupnoj točki i obrnuto. Koliko je AUTHENTICATION poruka računalo poslalo pristupnoj točki linksys_ses_24086, koja ima MAC adresu Cisco-Li_f5:ba:bb (00:18:39:f5:ba:bb) počevši od t=49?

15 poruka



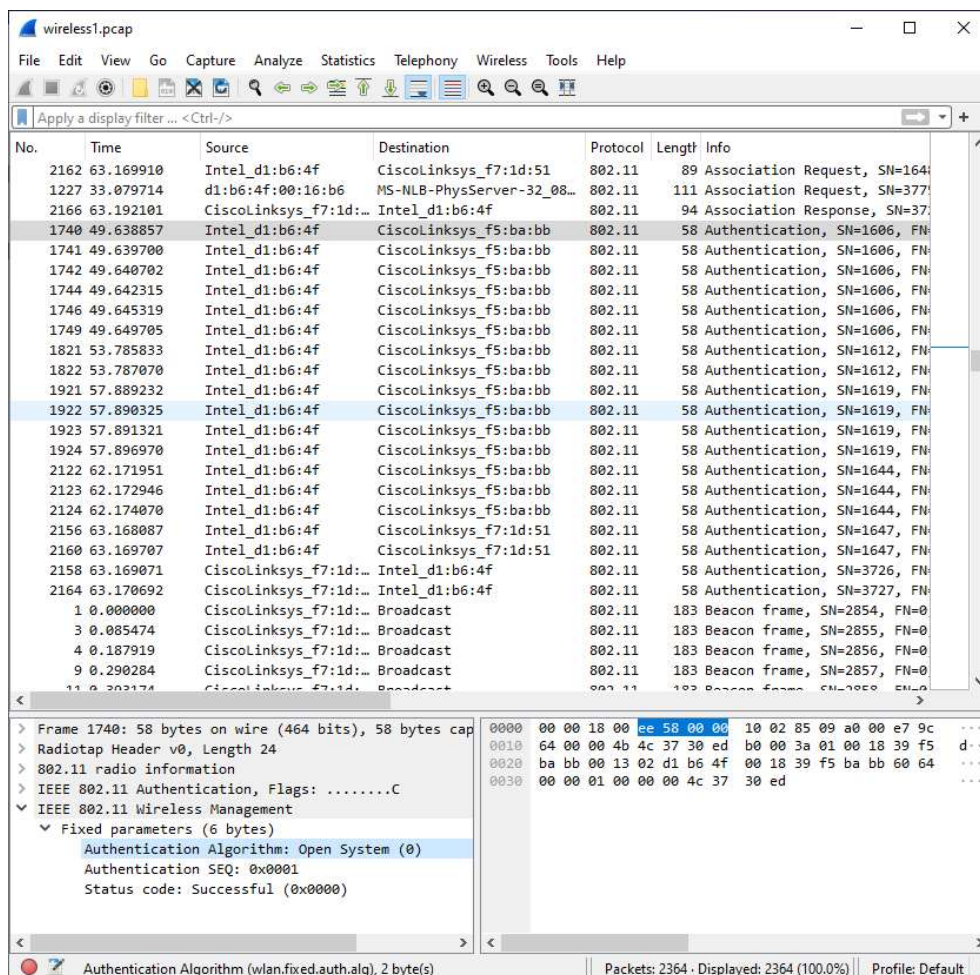
The image shows a Wireshark packet capture window titled 'wireless1.pcap'. The main pane displays a list of captured packets. The 'Info' pane on the right shows details for the selected packet (No. 2160). The status bar at the bottom indicates 'Ready to load or capture', 'Packets: 2364 · Displayed: 2364 (100.0%) · Selected: 17 (0.7%)', and 'Profile: Default'.

No.	Time	Source	Destination	Protocol	Length	Info
2162	63.169910	Intel_d1:b6:4f	CiscoLinksys_f7:1d:51	802.11	89	Association Request, SN=1644, FN=0
1227	33.079714	d1:b6:4f:00:16:b6	MS-NLB-PhysServer-32_08...	802.11	111	Association Request, SN=377, FN=0
2166	63.192101	CiscoLinksys_f7:1d:51	Intel_d1:b6:4f	802.11	94	Association Response, SN=37, FN=0
1740	49.638857	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1606, FN=0
1741	49.639700	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1606, FN=0
1742	49.640702	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1606, FN=0
1744	49.642315	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1606, FN=0
1746	49.645319	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1606, FN=0
1749	49.649705	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1606, FN=0
1821	53.785833	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1612, FN=0
1822	53.787070	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1612, FN=0
1921	57.889232	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1619, FN=0
1922	57.890325	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1619, FN=0
1923	57.891321	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1619, FN=0
1924	57.896970	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1619, FN=0
2122	62.171951	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1644, FN=0
2123	62.172946	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1644, FN=0
2124	62.174070	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	58	Authentication, SN=1644, FN=0
2156	63.168087	Intel_d1:b6:4f	CiscoLinksys_f7:1d:51	802.11	58	Authentication, SN=1647, FN=0
2160	63.169707	Intel_d1:b6:4f	CiscoLinksys_f7:1d:51	802.11	58	Authentication, SN=1647, FN=0
2158	63.169071	CiscoLinksys_f7:1d:51	Intel_d1:b6:4f	802.11	58	Authentication, SN=3726, FN=0
2164	63.170692	CiscoLinksys_f7:1d:51	Intel_d1:b6:4f	802.11	58	Authentication, SN=3727, FN=0
1	0.000000	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=2854, FN=0
3	0.085474	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=2855, FN=0
4	0.187919	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=2856, FN=0
9	0.290284	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=2857, FN=0
11	0.303174	CiscoLinksys_f7:1d:51	Broadcast	802.11	183	Beacon frame, SN=2858, FN=0

Slika 15 - Authentication

11. Želi li računalo autentikaciju koja zahtjeva korištenje kriptografskog ključa ili otvoren pristup?

Otvoren pristup (ako je vrijednost polja "Authentication Algorithm" 0, računalo želi otvoren pristup, a ako je vrijednost polja "Authentication Algorithm" 1, računalo želi autentikaciju koja zahtjeva korištenje kriptografskog ključa.



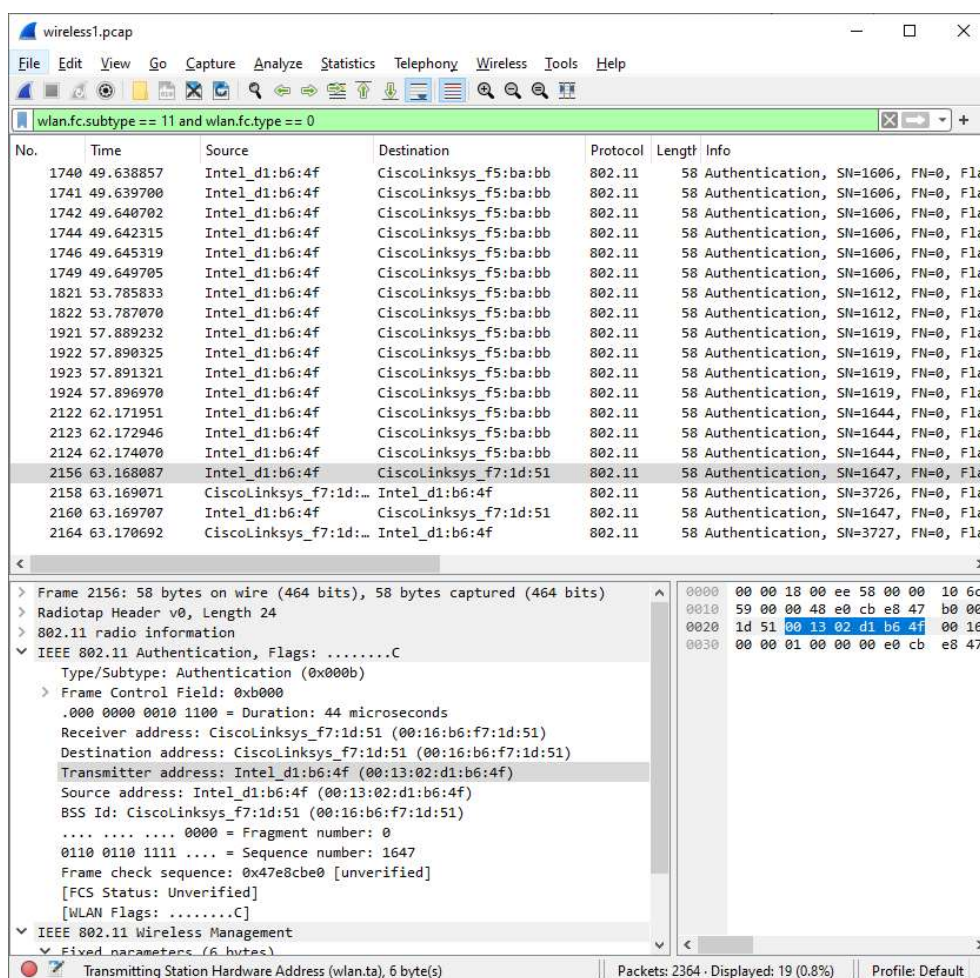
Slika 16 – otvoren pristup

12. Vidite li AUTHENTICATION odgovor od linksys_ses_24086 AP?

Ne

13. Što se događa kada računalo odustaje od povezivanja s linksys_ses_24086 AP, te se onda pokušava povezati s 30 Munroe St AP. Pogledajte AUTHENTICATION okvire koje računalo šalje AP i obrnuto. U kojem trenutcima računalo šalje AUTHENTICATION okvir 30 Munroe St. AP, a kada prima AUTHENTICATION odgovor od AP?

Računalo odustaje od povezivanja s `linksys\_ses\_24086` AP i šalje AUTHENTICATION okvir `30 Munroe St` AP u trenutku `t=63.168087`, a AUTHENTICATION odgovor od `30 Munroe St` AP prima neposredno nakon toga u trenutku `t=63.169071`.



Slika 17 - povezivanje s 30 Munroe St AP

14. Pronađite ASSOCIATE REQUEST koji računalo šalje pristupnoj točki i odgovarajući ASSOCIATE RESPONSE koje šalje pristupna točka računalu u postupku povezivanja. U kojem trenutku je poslan ASSOCIATE REQUEST za 30 Munroe St, a kada je stigao odgovarajući ASSOCIATE REPLY odgovor?

Request id e u t=63.169910

The screenshot displays the Wireshark network protocol analyzer interface. At the top, the title bar reads "wireless1.pcap". Below it is a menu bar with options: File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help.

A toolbar contains icons for common actions like opening files, saving, zooming, and filtering. Below the toolbar, a green filter bar contains the expression "wlan.fc.subtype < 2 and wlan.fc.type == 0".

The main packet list pane shows a series of captured packets:

No.	Time	Source	Destination	Protocol	Length	Info
1227	33.079714	d1:b6:4f:00:16:b6	MS-NLB-PhysServer-32_08..	802.11	111	Association Request, SN=3775, FN=0
1750	49.651078	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1607, FN=0
1751	49.653218	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1607, FN=0
1824	53.789944	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1613, FN=0
1825	53.790943	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1613, FN=0
1827	53.793568	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1613, FN=0
1926	57.903699	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1620, FN=0
1927	57.904945	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1620, FN=0
1932	57.911195	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1620, FN=0
1933	57.915945	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1620, FN=0
1934	57.924199	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1620, FN=0
1935	57.936216	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1620, FN=0
1937	57.939196	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1620, FN=0
2126	62.176945	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1645, FN=0
2127	62.178194	Intel_d1:b6:4f	CiscoLinksys_f5:ba:bb	802.11	107	Association Request, SN=1645, FN=0
2162	63.169910	Intel_d1:b6:4f	CiscoLinksys_f7:1d:51	802.11	89	Association Request, SN=1648, FN=0
2166	63.192101	CiscoLinksys_f7:1d:51	Intel_d1:b6:4f	802.11	94	Association Response, SN=3728, FN=0
2307	70.179949	CiscoLinksys_f5:ba:bb	f9:ff:ff:ff:ff:ff	802.11	132	Fragmented IEEE 802.11 frame

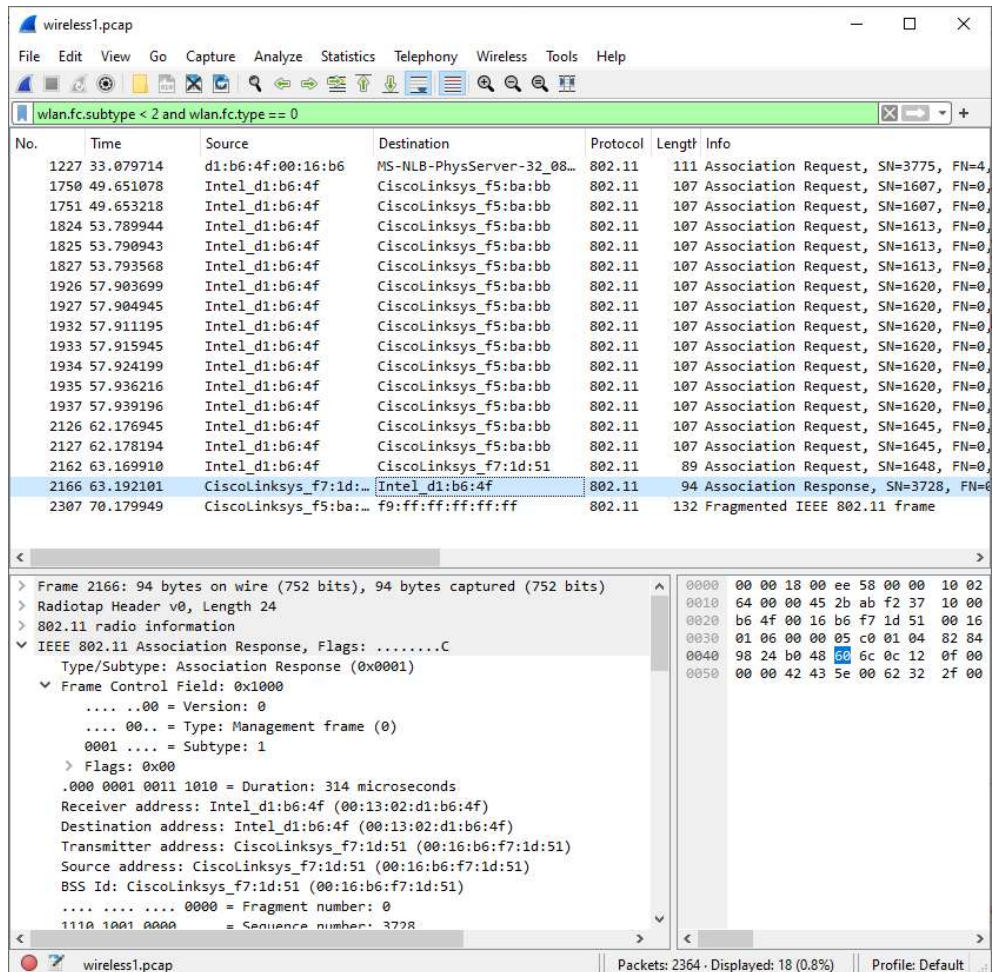
The bottom pane shows the detailed view of the selected packet (No. 2162). It displays the "Tagged parameters" section, which includes:

- Spectrum Management: Not Implemented
- QoS: Implemented
- Short Slot Time: In use
- Automatic Power Save Delivery: Implemented
- Radio Measurement: Not Implemented
- EPD: Not Implemented
- Reserved: 1
- Listen Interval: 0x000a

The "Tagged parameters" section also lists the SSID as "30 Munroe St" and provides information about supported rates and QoS capabilities.

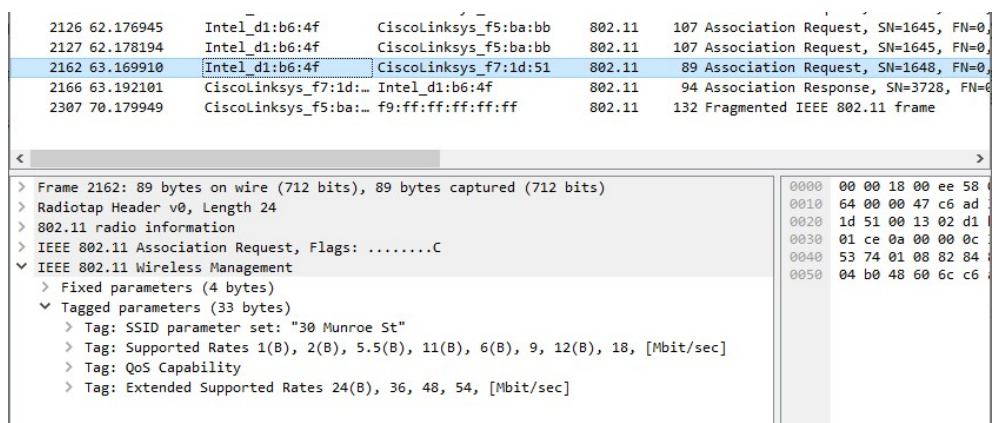
Slika 18 - REQUEST

Response ide u t=63.192101



Slika 19 - RESPONSE

15. Koje brzine prijenosa podržava 802.11 kartica računala, a koje pristupna točka?



Slika 20 – brzine 1

2162	63.169910	Intel_d1:b6:4f	CiscoLinksys_f7:1d:51	802.11	89 Association Request, SN=1648, FN=0
2166	63.192101	CiscoLinksys_f7:1d:51	Intel_d1:b6:4f	802.11	94 Association Response, SN=3728, FN=0
2307	70.179949	CiscoLinksys_f5:ba:5e	f9:ff:ff:ff:ff:ff	802.11	132 Fragmented IEEE 802.11 frame

> Frame 2166: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) > Radiotap Header v0, Length 24 > 802.11 radio information > IEEE 802.11 Association Response, Flags:C > IEEE 802.11 Wireless Management		0000 00 00 18 00 ee 58 0010 64 00 00 45 2b ab 0020 b6 4f 00 16 b6 f7 0030 01 06 00 00 05 c0 0040 98 24 b0 48 60 6c 0050 00 00 42 43 5e 00
--	--	--

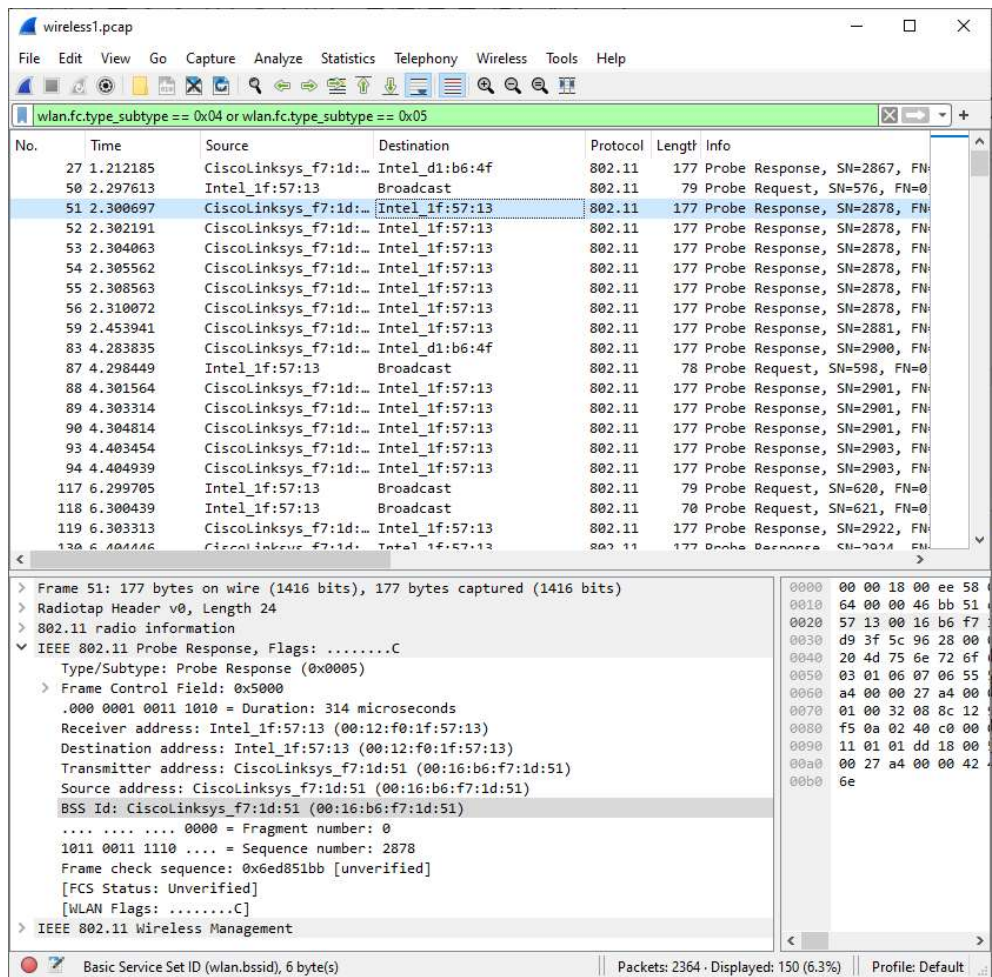
Slika 21 – brzine 2

16. Među snimljenim okvirima nalaze se PROBE REQUEST i PROBE RESPONSE okviri. Koje su MAC adrese pošiljatelja, primatelja i BSS ID kod ovih okvira? Koja je svrha ovog tipa okvira? (Odgovor možete naći u predloženim online materijalima).

No.	Time	Source	Destination	Protocol	Length	Info
27	1.212185	CiscoLinksys_f7:1d:51	Intel_d1:b6:4f	802.11	177	Probe Response, SN=2867, FN=0
50	2.297613	Intel_1f:57:13	Broadcast	802.11	79	Probe Request, SN=576, FN=0
51	2.300697	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2878, FN=0
52	2.302191	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2878, FN=0
53	2.304063	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2878, FN=0
54	2.305562	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2878, FN=0
55	2.308563	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2878, FN=0
56	2.310072	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2878, FN=0
59	2.453941	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2881, FN=0
83	4.283835	CiscoLinksys_f7:1d:51	Intel_d1:b6:4f	802.11	177	Probe Response, SN=2900, FN=0
87	4.298449	Intel_1f:57:13	Broadcast	802.11	78	Probe Request, SN=598, FN=0
88	4.301564	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2901, FN=0
89	4.303314	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2901, FN=0
90	4.304814	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2901, FN=0
93	4.403454	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2903, FN=0
94	4.404939	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2903, FN=0
117	6.299705	Intel_1f:57:13	Broadcast	802.11	79	Probe Request, SN=620, FN=0
118	6.300439	Intel_1f:57:13	Broadcast	802.11	70	Probe Request, SN=621, FN=0
119	6.303313	CiscoLinksys_f7:1d:51	Intel_1f:57:13	802.11	177	Probe Response, SN=2922, FN=0

> Frame 50: 79 bytes on wire (632 bits), 79 bytes captured (632 bits) > Radiotap Header v0, Length 24 > 802.11 radio information > IEEE 802.11 Probe Request, Flags:C Type/Subtype: Probe Request (0x0004) > Frame Control Field: 0x4000 .000 0000 0000 0000 = Duration: 0 microseconds Receiver address: Broadcast (ff:ff:ff:ff:ff:ff) Destination address: Broadcast (ff:ff:ff:ff:ff:ff) Transmitter address: Intel_1f:57:13 (00:12:f0:1f:57:13) Source address: Intel_1f:57:13 (00:12:f0:1f:57:13) BSS Id: Broadcast (ff:ff:ff:ff:ff:ff) 0000 = Fragment number: 0 0010 0100 0000 = Sequence number: 576 Frame check sequence: 0xa373c5ff [unverified] [FCS Status: Unverified] [WLAN Flags:C] > IEEE 802.11 Wireless Management		0000 00 00 18 00 ee 58 0010 5e 00 00 0e ff c5 0020 ff ff 00 12 f0 1f 0030 00 09 48 6f 6d 65 0040 16 0c 12 18 24 32
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Slika 22 - PROBE REQUEST

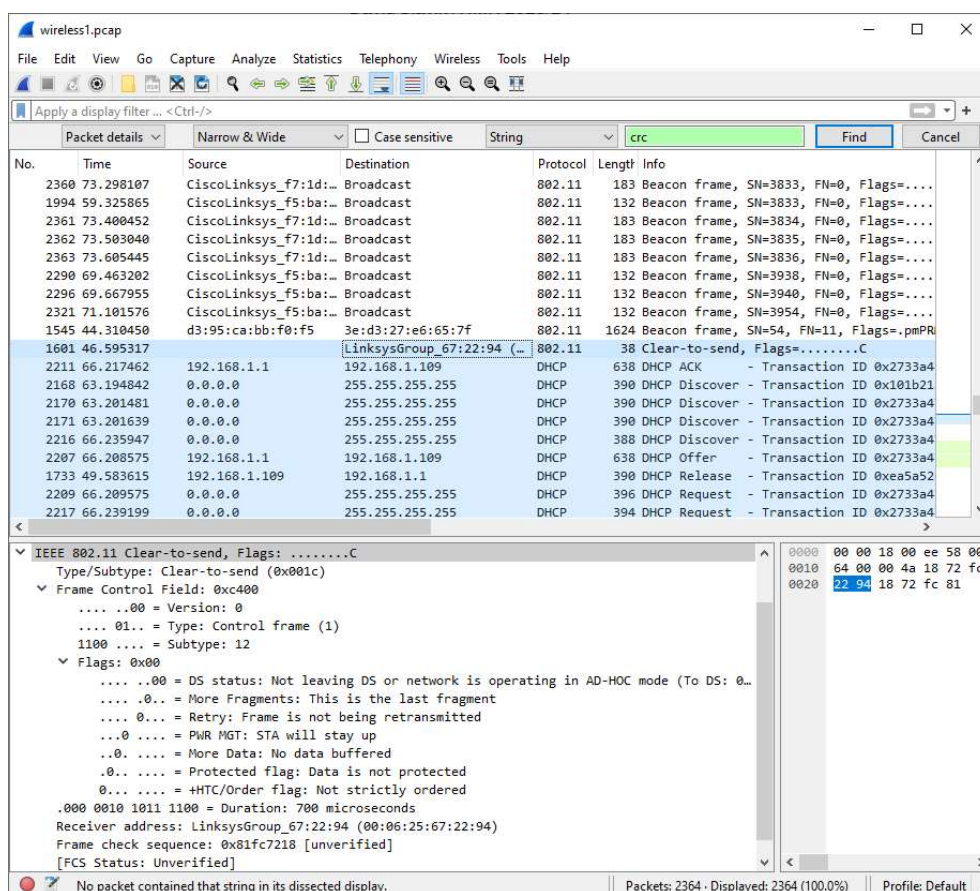


Slika 23 - PROBE RESPONSE

PROBE REQUEST: Ovaj okvir šalje uređaj koji traži dostupne pristupne točke (AP) u blizini. Uređaj šalje PROBE REQUEST kako bi otkrio koje mreže su dostupne.

PROBE RESPONSE: Ovaj okvir šalje pristupna točka (AP) kao odgovor na PROBE REQUEST. PROBE RESPONSE sadrži informacije o mreži, kao što su SSID, podržane brzine prijenosa i druge relevantne informacije.

17. Pronađite Clear to Send (CTS) okvir. Zašto se on koristi? Koliko MAC adresa koristi? Koje informacije sadrži?



Slika 24 – CTS okvir

Clear to Send (CTS) okvir se koristi u 802.11 mrežama za kontrolu pristupa mediju, smanjujući mogućnost sudara podataka. Kada uređaj želi poslati podatke, prvo šalje Request to Send (RTS) okvir, a ako je odobren, pristupna točka ili drugi uređaj šalje CTS okvir kao odgovor. CTS okvir koristi samo jednu MAC adresu jer je njegova svrha da obavijesti specifični uređaj (onaj koji je poslao RTS) da može započeti s prijenosom podataka, istovremeno informirajući sve ostale uređaje u mreži da trebaju odgoditi svoje prijenose. CTS okvir sadrži informacije o kontroli okvira, trajanju rezervacije medija, adresi primatelja i provjeri ispravnosti.

18. Koja je vrijednost zapisana u CRC polju CTS okvira? Zapišite tu vrijednost u obliku polinoma.

Frame check sequence -> $0x81fc7218_{16} = 10000001111111000111001000011000_2$